



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B.Sc Chemistry				
Course Title	INTRODUCTION TO POLYMER CHEMISTRY				
Type of Course	VOCATIONAL MINOR				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamental knowledge of chemistry principles and terminology. Understanding of basic organic chemistry concepts. Familiarity with chemical bonding and molecular structure. Proficiency in stoichiometry and reaction mechanisms.				
Course Summary	This course provides a comprehensive overview of polymer chemistry, covering fundamental concepts, properties, synthesis techniques, and industrial applications of polymers. Students explore polymer classification, properties, polymerization techniques, processing methods, and the characteristics of various commercial polymers. By the course's conclusion, students will have gained essential knowledge in polymer chemistry and its practical applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand various classification of polymers and types of polymerisation methods	U	C	Instructor-created exams / Quiz
CO2	Understand the important characteristics of polymers such as average molecular weight, glass transition temperature, viscoelasticity and degradation.	U	F	Class test /Assignment / Quiz
CO3	Appreciate the importance of processing techniques	U	F	Class test /Assignment / Quiz

CO4	Characterize different commercial polymers and to understand the significance of recycling	U	C	Class test /Assignment / Quiz
CO5	Develop practical skill in analyzing properties of polymers	Ap	An	viva/ practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours	Marks
I	Introduction to Polymers		15	32
	1	Polymers and macromolecules, Monomers, Homo and hetero polymers, Copolymers.	3	
	2	Classification based on origin (natural, semi-synthetic, and synthetic), Synthesis (addition and condensation), Structure (linear, branched chain, and cross-linked) intermolecular forces.	3	
	3	Elastomers, Fibers, Thermoplastics, and Thermosetting polymers, Tacticity.	3	
	4	Types of Polymerization: Chain and step growth polymerizations, Free radical, Ionic, and Coordination polymerizations with mechanism, and its advantages.	4	
	5	Ziegler-Natta polymerization (mechanism expected).	1	
	6	Ring-opening & Group transfer polymerization (Mechanism not needed).	1	
II	Properties of Polymers		10	22
	7	Molecular weights of polymers: Average molecular weights, Number average, Weight average, Sedimentation average (Method of determination not required).	2	
	8	Viscosity average molecular weight, Determination of viscosity average molecular weight.	2	
	9	Polydispersity index and Molecular weight distribution, Molecular weight and Degree of polymerization.	2	
	10	Glass transition temperature, Definition, Factors affecting Tg, Importance of Tg.	2	
	11	Viscoelasticity of polymers (Basic concepts only).	1	
	12	Polymer Degradation: Basic idea of thermal, photo, and oxidative degradation of polymers.	1	
III	Polymerisation Techniques		10	22

	13	Polymerization Techniques: Bulk, Solution, Suspension, Emulsion, Melt, Condensation, and Interfacial polycondensation polymerizations.	5	
	14	Polymer Processing: Calendering, Rotational molding, Compression, Injection molding, Blow molding, and Thermoforming.	5	
IV	Commercial Polymers		10	22
	15	Commercial Polymers: Preparation, Structure, Properties, and Applications of: Polyolefins (HDPE, LDPE, PP, and PS).	1	
	16	Vinyl polymers (PVC, PVP, and EVA, Saran), Fluoropolymers (Teflon), Acrylic polymers (PAN and PMMA).	1	
	17	Aliphatic polyamides (nylon 66 and nylon 6), Aromatic polyamides (kevlar).	1	
	18	Polyester (terylene), Polycarbonate (lexan), Polyurethanes.	2	
	19	Resins, Glyptal and Formaldehyde resins (UF, MF, and PF).	1	
	20	Rubbers (natural rubber - Vulcanization, EPDM, BR, SBR, Nitrile rubber, Neoprene, Butyl rubber, and Silicone rubber).	2	
	21	Conducting polymers, Doping (conduction mechanism not required).	1	
	22	Pollution due to plastics, Recycling of plastics, Plastic identification code.	1	
V	Practicals		30	
	Any Five of the following. 1. Identify everyday plastics by their physical properties 2. Determination of density of polymers 3. Effect of liquid on rubber 4. Determine glass and filler content 5. Determination of total solid content of latex 6. Determination of dry rubber content of latex 7. Determination of alkalinity of latex 8. Determination of KOH number Teacher can suggest determination of other properties of polymers other than mentioned above (Open ended)			

References:

1. Billmeyer Jr., F. W. (2007). *Textbook of Polymer Science*. John Wiley and Sons, New

Delhi.

- Gowariker, V. R. (2010). *Polymer Chemistry*. New Age International Pvt. Ltd., New Delhi.
- Sharma, B. K. (1989). *Polymer Chemistry*. Goel Publishing House, Meerut.
- Arora, M. G., Singh, M., & Yadav, M. S. (1989). *Polymer Chemistry, 2nd Revised Edn.* Anmolpublications Private Ltd., New Delhi.
- Saunders, K. J. (1988). *Organic Polymer Chemistry, 2nd Edn.* Chapman and Hall, London.
- Stevens, M. P. (1998). *Polymer Chemistry: An Introduction, 3rd Edn.* Oxford University Press, USA.
- Misra, G. S. (1993). *Introductory Polymer Chemistry*. New Age International, New Delhi.
- Bhatnagar, M. S. (2014). *Polymer Chemistry*. S Chand and Company Pvt. Ltd., New Delhi (Reprint).

Further reading:

- Seymour, R. B., & Carraher, C. E. (1981). *Polymer Chemistry: An Introduction*. Marcel Dekker, Inc. New York.
- Odian, G. (2004). *Principles of Polymerization, 4th Edn.* Wiley.
- Ghosh, P. (1991). *Polymer Science & Technology*. Tata McGraw-Hill Education.
- Lenz, R. W. (1967). *Organic Chemistry of Synthetic High Polymers*. Interscience Publishers, New York.
- Stevens, M. P. (1998). *Polymer Chemistry: An Introduction, 3rd Edn.* Oxford University Press.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	-	-	1	-	1	2	-	-	-	1	-	1
CO 2	2	1	-	1	2	2	2	-	1	-	1	-	2
CO 3	1	-	-	2	2	2	2	-	1	-	2	1	3
CO 4	2	-	1	3	2	3	3	-	1	1	2	3	3
CO 5	1	-	2	1	-	1	2	-	2	-	1	1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/seminar /viva	Practical skill Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5		✓	✓	